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METHOD FOR IMPLEMENTING AN AMMONIA-FUELLED FUEL CELL WITH DIHYDROGEN RECYCLING, AND ASSOCIATED PLANT

Abstract

A method for implementing an ammonia-burning fuel-cell system, which comprises: a) operating a fuel-cell unit; b) recovering a dinitrogen-and dihydrogen-rich anode gas stream; c) cooling the anode gas stream and condensing the water present in the gaseous anode stream to form a cooled gaseous anode stream; d) separating the cooled gaseous anode stream into a dinitrogen gas stream; and a dinitrogen-depleted anode stream; and e) injecting the dinitrogen-depleted anode stream; into the fuel cell unit so as to recycle the dinitrogen-depleted anode stream in the fuel cell unit.

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Background/Summary

[0001] The present invention relates to a method for implementing an ammonia-burning fuel-cell system, the method comprising a step for recycling dihydrogen formed during the method that is not consumed in the fuel cell.

[0002] The method of the invention is typically implemented in a fuel-cell plant enabling the production of an electric current by electrochemical reaction between dihydrogen H.sub.2 and dioxygen from the air. In the context of an ammonia-fueled fuel cell, the dihydrogen H.sub.2 used as the fuel fluid is generated during cell operation by the decomposition of ammonia NH.sub.3 at the fuel cell, in particular at the cell anode.

[0003] The fuel cell typically consists of one or more fuel cells, preferably operating at high temperature and enabling internal cracking of ammonia. Examples that may be cited include solid oxide fuel cells (SOFCs).

[0004] These batteries are currently designed mainly for stationary applications, with output power ranging from 1 kW to 2 MW.

[0005] An SOFC element is generally made up of four layers, three of which are ceramic. A single stack made up of these four superimposed layers is typically just a few millimeters thick. Dozens of these stacks are then superimposed in series to form a stack.

[0006] In these cells, oxygen ions formed on the cathode side are moved through a solid oxide used as a high-temperature electrolyte to react with the fuel gas, notably dihydrogen H.sub.2, on the anode side.

[0007] This electrochemical reaction leads to the production of electricity, as well as the formation of water, the by-product of the electrochemical reaction. Dinitrogen, resulting from the decomposition of ammonia, is also recovered at the stack outlet, as a mixture with the water formed.

[0008] In order to prevent and/or limit damage to the electrochemical cells during their operation, in particular due to the potential differences to which they are subjected, the flow rates of fuel, in particular ammonia, and oxidant, in particular air, are adjusted so that the amount of dihydrogen H.sub.2 in the cell always exceeds the maximum reaction capacity in the cell. As a result, a significant amount of unconsumed hydrogen is recovered from the SOFC element. The conversion rate of dihydrogen in a SOFC element is typically around 60%. In other words, around 40% of the hydrogen formed is recovered from the cell without having reacted with oxygen ions. Similarly, the air leaving the fuel cell is depleted in oxygen, but still contains a significant amount.

[0009] In known plants, the unconsumed dihydrogen is recovered at the stack outlet and then burned in a catalytic converter to be recovered in the form of thermal energy. One example is the method described in US 2021/0214849.

[0010] However, the use of dihydrogen by combustion is limited. In particular, it does not achieve electrical efficiencies as high as those theoretically permitted by the fuel cell. What's more, the operation of catalytic furnaces regularly leads to the formation of nitrogen oxides (also known as "NOx"), which can be difficult to manage. As nitrogen oxides are pollutants for the environment, adaptations to methods are necessary to prevent their release into the atmosphere.

[0011] US 2014/007889 describes a method for implementing an ammonia-fueled fuel-cell system wherein the anode output stream is fed into an aftertreatment unit wherein H.sub.2-dihydrogen is

separated from water and dinitrogen. The H.sub.2-dihydrogen stream recovered in this way is then injected into the fuel cell for recycling. H.sub.2-dihydrogen recovery is achieved by means of a series of hydrogen pumps enabling the recovery of a high-purity H.sub.2-dihydrogen stream. [0012] However, hydrogen pumps are expensive and bulky pieces of equipment. Their energy consumption is also high. In particular, the amount of energy required to pump the hydrogen is generally only slightly less than the amount of energy recovered by recycling the hydrogen. The energy gain associated with recycling dihydrogen using hydrogen pumps is negligible.

[0013] One aim of the invention is therefore to provide a method for implementing a fuel-cell system that enables the H.sub.2-dihydrogen present in the anode effluent to be recovered efficiently and reliably for recycling at the cell inlet, in particular with the aim of increasing the overall energy performance of the method, while limiting the investment and operating costs of the method.

[0014] In particular, the aim of the invention is to provide an integrated method which, by minimizing the energy requirement for H.sub.2-dihydrogen recovery, improves the overall energy performance of the electrical power generation unit.

[0015] Another aim of the invention is to provide a method for implementing an ammonia-burning fuel-cell system, the operation of which does not lead to the emission of nitrogen oxides (NO_x) in the exhaust gas.

[0016] To this end, the invention relates to a method for implementing a fuel-cell system, comprising the following steps: [0017] a) operating a fuel-cell unit comprising at least one anode system and at least one cathode system, said fuel-cell unit being continuously supplied with an ammonia-rich gas stream injected at the anode system and with a dioxygen-rich gas stream injected at the cathode system; [0018] b) recovering a dinitrogen-and dihydrogen-rich anode gas stream and a cathode gas stream at the outlet of the fuel-cell unit; [0019] c) cooling the anode gas stream and condensing the water present in the anode gas stream to form a cooled anode gas stream; [0020] d) separating the cooled anode gas stream into a dinitrogen gas stream and a dinitrogen-depleted anode stream; [0021] e) injecting at least part of the dinitrogen-depleted anode stream into the fuel-cell unit, in particular at the anode system, so as to recycle said at least part of the dinitrogen-depleted anode stream into the fuel-cell unit, [0022] wherein the dinitrogen-depleted anode stream has a residual dinitrogen content greater than or equal to 5% by volume. The method according to the invention may comprise one or more of the following features, considered alone or according to any technically possible combination: [0023] the dinitrogen-depleted anode stream has a dinitrogen content greater than 10% by volume, preferably greater than 20% by volume, more preferentially from 10% to 50% by volume, even more preferentially from 15% to 40% by volume, advantageously from 20 to 30% by volume; [0024] said at least part of the dinitrogen-depleted anode stream is injected into the ammonia-rich gas stream to form an ammonia-rich feed stream, said feed stream then being injected into the fuel-cell unit, in particular at the anode system; [0025] the feed stream has a dihydrogen content greater than or equal to 15% volume, preferably greater than or equal to 40% volume, more preferentially from 40% to 50% volume; [0026] step d) of separating the cooled anode gas stream is carried out by membrane separation, pressure swing adsorption, temperature swing adsorption, or pressure-temperature swing adsorption; [0027] the method comprises between steps c) and d) an intermediate step of compressing the cooled anode gas stream, preferably to a pressure greater than or equal to 10 bara, more preferentially greater than or equal to 20 bara, typically between 20 bara and 40 bara; [0028] the anode gas stream is at least partially cooled in step c) by heat exchange with the ammonia-rich gas stream or feed stream; [0029] the dioxygen-rich gas stream, before being introduced into the fuel-cell unit, is at least partially heated by heat exchange with the cathode gas stream; [0030] the method does not comprise a step of burning the anode gas stream, not even one of partial burning.

[0031] The invention further relates to a fuel-cell plant comprising: [0032] a fuel-cell unit comprising an inlet for introducing an ammonia-rich gas stream, an inlet for introducing a dioxygen-rich gas stream, an outlet for recovering an anode gas stream, and an outlet for

recovering a cathode gas stream, [0033] a cooling and condensing unit for cooling and drying the anode gas stream to form a cooled anode gas stream, [0034] a separation unit for separating the cooled anode gas stream into a dinitrogen gas stream and a dinitrogen-depleted anode stream, said separation unit comprising a dinitrogen-depleted anode stream recovery outlet, said dinitrogen-depleted anode stream recovery outlet being connected to the inlet for introducing the ammonia-rich gas stream into the fuel-cell unit, wherein the separation unit comprises at least one separation module selected from the group consisting of: a membrane separation unit, a pressure swing adsorption unit, a temperature swing adsorption unit, a pressure-temperature swing adsorption unit and any combination thereof. [0035] The plant according to the invention may comprise the feature that the separation unit further comprises a compression module, located upstream of said separation module.

Description

[0036] The invention will be better described upon reading the following description, given solely by way of example, and made with reference to the accompanying drawing, wherein:

[0037] FIG. 1 is a block diagram showing a plant for implementing a method according to the invention.

[0038] A plant **10** according to the invention is illustrated schematically in FIG. 1.

[0039] The plant **10** is designed to generate an electric current **12** from an ammonia gas stream **14** and a dioxygen-rich gas stream **16**.

[0040] The plant **10** is also designed to recover, as effluents, on the one hand a dinitrogen gas stream **18** and on the other hand a dioxygen-depleted gas stream **20**.

[0041] The plant **10** is connected upstream to a first storage unit **22** for storing ammonia gas **14**, and to a second storage unit **24** for storing dioxygen-rich gas **16**.

[0042] Alternatively, the storage unit **22** can be replaced by a gas distribution network for feeding ammonia gas **14** into the plant **10** of the invention.

[0043] When the dioxygen-rich gas **16** is air, the storage unit **24** can be replaced by a system of pump(s) and compressor(s) for feeding atmospheric air, possibly compressed, into the plant **10** of the invention.

[0044] The ammonia gas stream **14** is predominantly made up of ammonia. Preferably, the ammonia gas stream **14** has an ammonia content greater than or equal to 90% by volume, more preferentially greater than or equal to 95% by volume, even more preferentially greater than or equal to 97% by volume, advantageously greater than or equal to 98% by volume, more advantageously greater than or equal to 99% by volume.

[0045] Preferably, ammonia gas **14** consists essentially of ammonia, and possibly impurities.

[0046] In one embodiment, the ammonia gas stream **14** may comprise very small quantities of impurities, in particular water and traces of oil.

[0047] Preferably, the ammonia gas stream **14** has an impurity content of less than or equal to 10% by volume, more preferentially less than or equal to 5% by volume, typically less than or equal to 1% by volume.

[0048] Preferably, the ammonia gas stream **14** is at a pressure ranging from 1 atm to 5 bara (absolute bar), more preferentially from 1 atm to 2 bara (absolute bar)

[0049] The dioxygen-rich gas stream **16** typically has a dioxygen content of at least 10% by volume, preferably 15% to 25% by volume of dioxygen. Advantageously, the dioxygen-rich gas stream **16** is an air stream.

[0050] Preferably, the dioxygen-rich gas stream **16** is at a pressure ranging from 1 atm to 5 bara (absolute bar).

[0051] The plant **10** is connected downstream to a plant **26** for collecting the dinitrogen gas stream

18, and to a plant **28** for collecting the dioxygen-depleted gas stream **20** for aftertreatment.

Alternatively, the dinitrogen gas stream **18** and/or the dioxygen-depleted gas stream **20** are released directly into the atmosphere.

[0052] Preferably, the dioxygen-depleted gas stream **20** is discharged directly into the atmosphere at the plant **28**.

[0053] Alternatively, the dioxygen-depleted gas stream **20** is collected in the plant **28** and routed to another plant for consumption or processing.

[0054] The plant **10** typically comprises a fuel-cell unit **30**, a cooling and condensation unit **32**, and a separation unit **34**.

[0055] Prior to its introduction into the fuel-cell unit **30**, the ammonia gas stream **14** is mixed with a recycling stream **36** recovered at the outlet of the separation unit **34** to form an ammonia-rich feed stream **38**. The nature of recycling stream **36** is described in detail below.

[0056] The fuel-cell unit **30** is designed to produce the electric current **12** from the feed stream **38** and the dioxygen-rich gas stream **16** to form an anode gas stream **40** and a cathode gas stream **42**.

[0057] The cooling and condensation unit **32** is designed to cool the anode gas stream **40** and extract at least some of the water by condensation to form a cooled anode gas stream **44**.

[0058] Finally, the separation unit **34** is designed to recover at least some of the dinitrogen present in the cooled anode gas stream **44** in the form of the dinitrogen gas stream **18**, and to recover a dinitrogen-depleted anode stream **36**, also referred to as the recycling stream **36**.

[0059] Preferably, the plant **10** according to the invention comprises, upstream of the fuel-cell unit **30**, a first heat exchanger system **46** designed to heat the ammonia-rich feed stream **38** prior to its introduction into the fuel-cell unit **30**.

[0060] For the purposes of this invention, a heat exchanger system comprises at least one heat exchanger. A heat exchanger system can thus comprise a single heat exchanger or a plurality of heat exchangers associated with each other.

[0061] The first heat exchanger system **46** thus comprises an inlet **48** for introducing the ammonia-rich feed stream **38** and an outlet **50** for recovering a heated fuel stream **52**.

[0062] Preferably upstream of the fuel-cell unit, the plant **10** further comprises a unit **54** for pretreating the dioxygen-rich gas stream **16**. The pretreatment unit **54** is designed on the one hand to compress and/or dry and/or filter, preferably compress, dry and filter, the dioxygen-rich gas stream **16**, and on the other hand to heat the dioxygen-rich gas stream **16** prior to its introduction into the fuel-cell unit **30**.

[0063] The pretreatment unit **54** firstly comprises a module **55** designed to compress and/or dry and/or filter, preferably compress, dry and filter, the dioxygen-rich gas stream **16** to form a pressurized dioxygen-rich gas stream **56**. The module **55** thus comprises an inlet **57** for introducing dioxygen-rich gas stream **16** and an outlet **58** for recovering pressurized dioxygen-rich gas stream **56**.

[0064] Preferably, the pretreatment unit **54** further comprises a second heat exchanger system **60** designed to heat the pressurized dioxygen-rich gas stream **56** prior to its introduction into the fuel-cell unit **30**. The second heat exchanger system **60** thus comprises an inlet **62** for introducing the pressurized dioxygen-rich gas stream **56** and an outlet **64** for recovering a heated dioxygen-rich stream **66**.

[0065] The fuel-cell unit **30** typically comprises at least one anode system **70** and at least one cathode system **72**, separated from each other by at least one electrolyte (not shown), in particular a solid electrolyte.

[0066] For the purposes of this invention, an anode (or cathode) system comprises at least one anode (or cathode). An anode (or cathode) system can thus comprise a single anode (or cathode) or a plurality of anodes (or cathodes) associated with each other.

[0067] The fuel-cell unit **30** comprises an inlet **72** for introducing the heated fuel stream **52**. The inlet **72** is designed to bring the heated fuel stream **52** into contact with the anode system **68**.

[0068] The fuel-cell unit **30** further comprises an inlet **74** for introducing the heated dioxygen-rich stream **66**. In particular, the inlet **74** is designed to bring the heated dioxygen-rich stream **66** into contact with the cathode system **70**.

[0069] The fuel-cell unit **30** further comprises an outlet **76** for recovering the anode gas stream **40**, which results from the operation of the fuel-cell unit **30** and is recovered from the anode system **68**.

[0070] The fuel-cell unit **30** further comprises an outlet **78** for recovering a dioxygen-depleted cathode gas stream **42**, which results from the operation of the fuel-cell unit **30** and recovered from the cathode system **70**.

[0071] The outlet **78** for recovering the dioxygen-depleted cathode gas stream **42** is connected to the plant **28**, preferably via the pretreatment unit **54**.

[0072] In one embodiment, the outlet **78** is connected to the pretreatment unit **54**, which in turn is connected to the plant **28**, so that the dioxygen-depleted cathode gas stream **42** undergoes one or more aftertreatments before being transferred to the plant **28**.

[0073] Preferably, the outlet **78** for recovering the cathode gas stream **42** is connected upstream of the plant **28** to the second heat exchanger system **60**, so that the dioxygen-depleted cathode gas stream **42** is cooled there, by heat exchange with the pressurized dioxygen-rich gas stream **56**, to form the dioxygen-depleted gas stream **20**. The second heat exchanger system **60** then comprises an inlet **79** for introducing the cathode gas stream **42** and an outlet **80** for recovering the dioxygen-depleted gas stream **20**.

[0074] In an even more preferred embodiment, the outlet **80** of the second heat exchanger system **60** is connected to a supplementary cooling system **81** designed to further cool the dioxygen-depleted gas stream **20**, before transferring it to the plant **28**. The supplementary cooling system **81** then comprises an inlet **82A** for introducing the dioxygen-depleted gas stream **20** and an outlet **82B** for recovering a cooled dioxygen-depleted gas stream **83**. The outlet **82B** of the supplementary cooling system **81** is directly connected to the plant **28**.

[0075] The supplementary cooling system **81** may contain heat exchangers that enable the recovered heat to be used, for example, to supply a Rankine-type steam or electrical generation system, or a heat distribution network. Preferably, when there is no thermal integration for heat recovery, the supplementary cooling system **81** consists of coolers such as air coolers or water exchangers to achieve the required cooling level.

[0076] This method is advantageous in that it enables the dioxygen-rich gas stream **16** to be heated by thermal integration before being introduced into the fuel-cell unit **30**, without the need for external energy input.

[0077] The cooling and condensation unit **32** typically comprises, on the one hand, a heat exchanger system **46** and a supplementary cooling module **84** designed to cool the anode gas stream **40** and at least partially condense the water present in the anode gas stream **40** resulting from the operation of the fuel-cell unit **30**, and, on the other hand, a condensate recovery system **85**.

[0078] In one embodiment, the outlet **76** for recovering the anode gas stream **40** is connected to the first heat exchanger system **46** so that the anode gas stream **40** is cooled there by heat exchange with the ammonia-rich feed stream **38**, to form a pre-cooled anode stream **86**. The first heat exchanger system **46** thus comprises an inlet **88** for introducing the anode gas stream **40** and an outlet **89** for recovering the pre-cooled anode stream **86**.

[0079] Thus, according to this embodiment, the heat exchanger system of the cooling and condensation unit **32** is constituted by the first heat exchanger system **46** described above.

[0080] This method is advantageous in that it enables the feed stream **38** to be heated before being introduced into the fuel-cell unit **30**, and to cool the anode gas stream **40** before being introduced into the condensate recovery system **85**, by thermal integration without the need for external energy input.

[0081] In an alternative embodiment (not shown), the heat exchanger system of the cooling and

condensation unit **32** is separate from the first heat exchanger system **46**.

[0082] The supplementary cooling module **84** is designed to further cool the pre-cooled anode stream **86** exiting the first heat exchanger system **46** before it is introduced into the condensate recovery system **85**. In particular, the supplementary cooling module **84** cools the water present in the pre-cooled anode stream **86** to a temperature low enough to allow condensation.

[0083] The supplementary cooling module **84** thus comprises an inlet **90** for introducing the pre-cooled anode stream **86** and an outlet **91** for recovering a cooled anode stream **92**.

[0084] In a preferred embodiment, the supplementary cooling module **84** consists of a heat exchanger for recovering, preferably recovering and reusing, the heat from the pre-cooled anode stream **86**. This heat is recovered, for example, by feeding a steam generation system, a Rankine-type electrical generation system, or a heat distribution network.

[0085] Preferably, the supplementary cooling module **84** comprises one or more coolers, in particular selected from air coolers, water exchangers and any combination thereof.

[0086] The condensate recovery system **85** is designed to condense the water present in the cooled anode stream **92** to form a liquid water condensate **93** at the bottom of the condensate recovery system **85**, and the cooled anode gas stream **44** at the top of the condensate recovery system **85**. The condensate recovery system **85** thus comprises an inlet **94** for introducing the cooled anode stream **92**, an outlet **96** for recovering condensate **93**, and an outlet **98** for recovering the cooled anode gas stream **44**.

[0087] The plant **10** further comprises a system **99** for collecting liquid-water condensate **93**.

[0088] The separation unit **34** typically comprises a compression unit **100** and a separation module **102**.

[0089] The compression unit **100** is designed to at least partially compress the cooled anode gas stream **44** recovered at the outlet **98** of the cooling and condensation unit **32**. The compressor unit **100** thus comprises an inlet **104** for introducing the cooled anode gas stream **44** and an outlet **106** for recovering a compressed cooled anode gas stream **108**.

[0090] The nature of the compression unit **100** is not particularly limited. Any device known to the person skilled in the art that is capable of compressing the cooled anode gas stream **44** sufficiently for separation can be used. The compressor unit **100** consists, for example, of a series of compressors and/or blowers.

[0091] The separation module **102** is designed to partially extract the dinitrogen present in the compressed, cooled anode gas stream **108**, so as to recover the dinitrogen gas stream **18** on the one hand, and the dinitrogen-depleted anode stream **36** (also known as the recycling stream **36**) on the other. The separation module **102** thus comprises an inlet **110** for introducing the compressed, cooled anode gas stream **108**, an outlet **112** for recovering the dinitrogen gas stream **18**, and an outlet **114** for recovering the dinitrogen-depleted anode stream **36**.

[0092] The separation module **102** is typically made up of any equipment known to the person skilled in the art and capable of at least partially extracting the dinitrogen present in a gaseous mixture of dihydrogen and dinitrogen.

[0093] The separation module **102** can optionally be supplemented by one or more additional items of equipment chosen from the group of: a purification unit (not shown) to remove any nitrogen oxides (NO_x), a drying device and a superheater. When present, this additional equipment is typically placed upstream of the separation module **102**, typically between the compression unit **100** and the separation module **102**.

[0094] The separation module **102** typically comprises a membrane separation unit, a pressure-swing adsorption (PSA) unit, a temperature-swing adsorption (TSA) unit, a pressure-temperature-swing adsorption (PTSA) unit, or any combination thereof.

[0095] This equipment is well known to the person skilled in the art and will not be described further below.

[0096] Preferably, the outlet **112** for recovering the dinitrogen gas stream **18** is connected to the

plant **26** so that the dinitrogen gas stream **18** can be collected there for aftertreatment.

[0097] Preferably, the outlet **114** for recovering the dinitrogen-depleted anode stream **36** is connected to the inlet **48** of the first heat exchanger system **46** for mixing with the ammonia gas stream **14**.

[0098] According to a preferred embodiment, the plant **10** according to the invention does not comprise hydrogen pumps. Such equipment is expensive and bulky. They also require a significant amount of energy to operate, generally only slightly less than that recovered by recycling the hydrogen. The system according to the invention is therefore advantageous in that it dispenses with this costly, cumbersome, energy-intensive equipment.

[0099] The streams described in the plant are conflated with the pipes that carry them.

[0100] The implementation of a first method according to the invention will now be described.

[0101] The method according to the invention comprises firstly supplying at least one ammonia gas stream **14**, from the plant **22** (or a gas pipe network), and at least one dioxygen-rich gas stream **16**, from the plant **24**.

[0102] The ammonia gas stream **14** is first mixed with the dinitrogen-depleted anode stream **36** (also known as the recycling stream **36**) recovered at the outlet **114** of the separation unit **34** to form the ammonia-rich feed stream **38**.

[0103] As explained in detail below, the hydrogen not consumed by the fuel-cell unit **30** is recovered and reinjected upstream of the fuel-cell unit **30** in the form of the dinitrogen-depleted anode stream **36** for recycling.

[0104] The ratio at which the ammonia gas stream **14** and the dinitrogen-depleted anode stream **36** are mixed depends on the conversion rate of the fuel-cell unit **30**, which is typically 60%.

[0105] Preferably, the ammonia gas stream **14** and the dinitrogen-depleted anode stream **36** are mixed in a volume ratio ranging from 1:10 to 10:1, more preferentially from 5:10 to 1:1, advantageously from 1:2 to 4:5, typically 7:10.

[0106] Preferably, the ammonia-rich feed stream **38** has an ammonia NH_3 content of 15% to 75% by volume, more preferentially 30% to 60% by volume, advantageously 40% to 50% by volume, typically 45% by volume.

[0107] Preferably, the ammonia-rich feed stream **38** has a dihydrogen H_2 content of 15% to 75% by volume, more preferentially 30% to 60% by volume, advantageously 40% to 50% by volume, typically 45% by volume.

[0108] Preferably, the ammonia-rich feed stream **38** has a dinitrogen N_2 content of 0% to 50% by volume, more preferentially 5% to 30% by volume, advantageously 10% to 20% by volume, typically 10% by volume.

[0109] The ammonia-rich feed stream **38** is then introduced into the first heat exchanger system **46**, where it is heated to form a heated fuel stream **52**.

[0110] Preferably, upstream of the first heat exchanger system **46**, the ammonia-rich feed stream **38** has a temperature of 5° C. to 80° C., more preferentially 20° C. to 50° C., advantageously 30° C. to 40° C., typically 35° C.

[0111] Preferably, the ammonia-rich feed stream **38** is heated at the first heat exchange system **46** to a temperature lower than the temperature of the fuel-cell unit **30**, more preferentially to a temperature lower by 10° C. to 300° C. than that of the fuel-cell unit **30**, more preferentially lower by 50° C. to 200° C., typically lower by 100° C. to 150° C.

[0112] The heated fuel stream **52** is then introduced into the fuel-cell unit **30**.

[0113] At the same time, the dioxygen-rich gas stream **16** is introduced into the pretreatment unit **54**, where it is compressed and/or dried and/or filtered, preferably compressed, dried, and filtered, and also heated before being introduced into the fuel-cell unit **30**.

[0114] More particularly, the dioxygen-rich stream **16** is first introduced into the module **55** to be compressed and/or dried and/or filtered, preferably compressed, dried, and filtered, to form a pressurized dioxygen-rich gas stream **56** at the outlet.

[0115] The pressurized dioxygen-rich gas stream **56** is then introduced into the second heat exchanger system **60**, where it is heated to form the heated dioxygen-rich stream **66** at the outlet **64**.

[0116] Preferably, upstream of the second heat exchanger system **60**, the dioxygen-rich pressurized gas stream **56** has a temperature of 10° C. to 200° C., more preferentially 50° C. to 150° C., typically 75° C. to 120° C., typically 90° C. to 95° C.

[0117] Preferably, the dioxygen-rich pressurized gas stream **56** is heated at the second heat exchange system **60** to a temperature lower than that of the fuel-cell unit **30**, more preferentially to a temperature lower by 10° C. to 300° C. than the temperature of the fuel-cell unit **30**, more preferentially lower by 50° C. to 200° C., typically lower by 100° C. to 150° C.

[0118] The heated dioxygen-rich stream **66** is then introduced into the fuel-cell unit **30**.

[0119] The dioxygen-rich stream **16** and the feed stream **38** are pre-heated to maintain a relatively constant temperature inside the stack unit **30** and thus guarantee optimum performance.

[0120] The fuel-cell unit **30**, continuously supplied with the heated fuel stream **52** and the heated dioxygen-rich stream **66**, is then put into operation.

[0121] To do this, the heated fuel stream **52** is introduced into the fuel-cell unit **30** at the inlet **72**. In particular, the heated fuel stream **52** is injected into the anode system **68**.

[0122] The heated dioxygen-rich stream **66** is introduced into the inlet **74** of the fuel-cell unit **30**. In particular, the heated dioxygen-rich stream **66** is injected into the cathode system **70**.

[0123] On entering the fuel-cell unit **30**, the heated fuel stream **52** undergoes a first catalytic cracking step, during which the ammonia present in the heated fuel stream **52** decomposes into dinitrogen gas N_{2} and hydrogen gas H_{2} . The catalytic cracking of ammonia is initiated by the high temperature inside the fuel-cell unit **30** in the presence of the catalytic materials in the anode system **68**. The temperature inside the fuel-cell unit **30** during operation depends on the type of high-temperature fuel cell and the operating choices made. In the case of an SOFC fuel cell, this is typically 500° C. to 1100° C., preferably 700° C. to 900° C.

[0124] Preferably, all the ammonia in the heated fuel stream **52** is decomposed into dinitrogen gas N_{2} and hydrogen gas H_{2} at the fuel-cell unit (**30**).

[0125] On contact with the anode system **68**, H_{2} hydrogen dissociates into H^{+} protons and electrons. The electrons thus released accumulate at the anode system **68** and are transferred to the cathode system **70** of the fuel-cell unit **30** by means of an external circuit, the movement of electrons from the anode system **68** to the cathode system **70** generating the electric current **12**.

[0126] The dinitrogen N_{2} present in the fuel-cell unit **30**, particularly in the anode system **68**, is inert at the operating temperatures of the fuel-cell unit **30**. In particular, the nitrous oxide $N_{2}O$ is not oxidized and ends up entirely in the anode gas stream **40** recovered at the outlet of the fuel-cell unit **30**.

[0127] In the cathode system **70**, the electrons carried by the electric current **12** are fixed by the dioxygen O_{2} present in the heated dioxygen-rich stream **66**. The dioxygen molecules O_{2} are then transformed into superoxide ions O_{2}^{-} by the capture of two electrons.

[0128] The superoxide ions O_{2}^{-} thus formed then pass through the electrolyte to the anode system **68**, where they react with the protons H^{+} generated earlier to form water $H_{2}O$.

[0129] The anode gas stream **40** and the cathode gas stream **42** are then recovered at the outlet of the fuel-cell unit **30**.

[0130] The cathode gas stream **42**, corresponding to the residual gas stream recovered at the outlet of cathode system **70**, is recovered at the outlet **78** of the fuel-cell unit **30**. The stream **42** has a reduced dioxygen content compared to the initial heated dioxygen-rich stream **66**.

[0131] Typically, the cathode gas stream **42** has a dioxygen content of less than or equal to 20% by mole, preferably from 5% to 15% by mole.

[0132] The cathode gas stream **42** is then conveyed to the plant **28**.

[0133] In one embodiment, before reaching the plant **28**, the cathode gas stream **42** is cooled by

passing through the second heat exchanger system **60**. In this way, the cathode gas stream **42** is cooled by heat exchange with the pressurized dioxygen-rich gas stream **56** to form the dioxygen-depleted gas stream **20**.

[0134] The cathode gas stream **42** recovered at the outlet **78** of the fuel-cell unit **30** has a temperature close to the temperature of the fuel-cell unit **30**.

[0135] Thus, in the case of an SOFC fuel cell, the cathode gas stream **42** recovered at the outlet **78** of the fuel-cell unit **30** typically has a temperature of 500° C. to 1100° C., preferably 700° C. to 900° C.

[0136] Preferably, the cathode gas stream **42** is cooled to a temperature ranging from 50° C. to 250° C., preferably from 100° C. to 200° C.

[0137] This method is advantageous in that it enables the dioxygen-rich gas stream **16** at the inlet to be heated, and the cathode gas stream **42** at the outlet to be cooled, by thermal integration without the need for external energy input.

[0138] According to an even more preferred method, the cathode gas stream **42** is cooled in at least two stages before being transferred to the plant **28**. In particular, the cathode gas stream **42** is first cooled in the second heat exchanger system **60** to an intermediate temperature of 200° C. to 350° C., preferably 250° C. to 300° C. Then, the dioxygen-depleted stream **20** recovered at the outlet **80** of the second heat exchanger system **60** is transferred to a supplementary cooling system **81** for further cooling of the cathode gas stream **42** to a temperature compatible with the characteristics of the plant **28** for collecting the dioxygen-depleted stream **20**. The cooled, dioxygen-depleted gas stream **83** recovered at the outlet **82B** of the supplementary cooling system **81** typically has a temperature below 200° C., preferably between 100° C. and 200° C., advantageously around 150° C. This supplementary cooling system **81** may contain heat exchangers enabling the recovered heat to be valorized by feeding, for example, a Rankine-type steam or electrical generation system, or a heat distribution network. Preferably, the supplementary cooling system **81** consists of coolers such as air coolers or water exchangers to achieve the required cooling level

[0139] The anode gas stream **40**, corresponding to the residual gas stream recovered at the outlet of the anode system **68**, is recovered at the outlet **76** of the fuel-cell unit **30**. The anode gas stream **40** is made up of a mixture of water H_2O resulting from the operation of the fuel-cell unit **30**, as well as unreacted dinitrogen N_2 and dihydrogen H_2 .

[0140] Preferably, at most 40% by volume of the dihydrogen H_2 formed by ammonia decomposition inside the fuel-cell unit **30** is recovered in the anode gas stream **40**, more preferentially from 20% to 30% by volume.

[0141] Preferably, the anode gas stream **40** has a dinitrogen content of 10% to 40% by volume, more advantageously 20% to 30% by volume, typically 25% by volume.

[0142] Preferably, the anode gas stream **40** has a water H_2O content of 30% to 60% by volume, more preferentially 40% to 50% by volume, typically 45% by volume.

[0143] In the case of an SOFC fuel cell, the anode gas stream **40** recovered at the outlet **76** of the fuel-cell unit **30** typically has a temperature of 500° C. to 1100° C., typically 700° C. to 900° C.

[0144] The entire anode gas stream **40** is then introduced into the cooling and condensation unit **32**, where it is cooled and at least some, preferably most, of the water formed is removed.

[0145] The anode gas stream **40** is first cooled by passing through one or more heat exchanger systems.

[0146] Preferably, the anode gas stream **40** is first introduced into the first heat exchanger system **46**, where it is cooled by heat exchange with the ammonia-rich feed stream **38**. A pre-cooled anode stream **86** is recovered at the outlet **89** of the first heat exchanger system **46**.

[0147] In the case of an SOFC fuel cell, the anode gas stream **40** recovered at the outlet **76** of the fuel-cell unit **30** typically has a temperature of 500° C. to 1100° C., preferably 700° C. to 900° C.

[0148] Preferably, the anode gas stream **40** is cooled in the first heat exchanger system **46** to a temperature of 300° C. to 500° C., preferably 350° C. to 450° C.

[0149] Preferably, the pre-cooled anode stream **86** is then introduced into the supplementary cooling module **84** for further cooling. A cooled anode stream **92** is recovered from the outlet **91** of the supplementary cooling module **84**.

[0150] Preferably, the pre-cooled anode stream **86** is cooled in the supplementary cooling module **84** to a temperature of 100° C. or less, more preferentially between 10° C. and 50° C., typically between 30° C. and 40° C.

[0151] The cooled anode stream **92** is then introduced into the condensate recovery system **85** to remove the water present in the cooled anode stream **92**.

[0152] In particular, the cooled anode stream **92** is introduced into the condensate recovery system **85**. A liquid water condensate **93** is collected at the bottom of the condensate recovery system **85**, while the cooled anode gas stream **44** is collected at the top of the condensate recovery system **85**.

[0153] The condensate **93** is then sent to plant **99** for aftertreatment.

[0154] Preferably, if the fuel-cell system is operates at near-atmospheric pressure, the cooled anode gas stream **44** has a water content of less than or equal to 10% by volume, more preferentially less than or equal to 6% by volume.

[0155] The entire cooled anode gas stream **44** is then injected into the separation unit **34**, where it is split into the dinitrogen gas stream **18** and the dinitrogen-depleted anode stream **36**, also known as the recycling stream **36**.

[0156] Preferably, the cooled anode gas stream **44** undergoes a preliminary compression step. To achieve this, the cooled anode gas stream **44** is fed into the compressor unit **100** to form the compressed cooled anode gas stream **108**.

[0157] Preferably, if the fuel-cell system is operated at near-atmospheric pressure, the cooled anode gas stream **44** has a pressure of 0.7 bara to 2 bara (bar absolute), more preferentially 1 bara to 1.5 bara (bar absolute).

[0158] Preferably, the cooled, anode gas stream **44** is compressed in the compressor unit **100** to a pressure ranging from 10 bara to 100 bara (bar absolute), more preferentially from 20 bara to 40 bara (bar absolute). The final pressure of the cooled gas stream **44** depends on the characteristics of the separation module **102**

[0159] The compressed, cooled anode gas stream **108** is then introduced into the separation module **102** to form the dinitrogen gas stream **18** on the one hand, and the dinitrogen-depleted anode stream **36** (recycling stream **36**) on the other.

[0160] The dinitrogen gas stream **18** is then sent to the plant **26** for aftertreatment.

[0161] Preferably, the dinitrogen gas stream **18** consists essentially of ammonia, and possibly impurities.

[0162] More preferentially, the dinitrogen gas stream **18** has a dinitrogen content greater than or equal to 99.00% by volume, more preferentially between 99.90% and 99.99% by volume.

[0163] In one embodiment, the dinitrogen gas stream **18** may comprise very small quantities of impurities.

[0164] Impurities that may be present in the dinitrogen gas stream **18** include dihydrogen, ammonia, water, nitrogen oxides (or NO_x) and traces of oil. The presence of a purification and/or drying unit, particularly upstream of the separation unit **102**, reduces the impurity content.

[0165] Preferably, the dinitrogen gas stream **18** has an impurity content of less than or equal to 10% by volume, more preferentially less than or equal to 5% by volume, typically less than or equal to 1% by volume.

[0166] Preferably, the dinitrogen-depleted anode stream **36** (recycling stream **36**) has a dihydrogen H₂ content greater than or equal to 50% by volume, more preferentially greater than or equal to 70% by volume, typically between 70% and 90% by volume, for example between 70% and 80% by volume.

[0167] Preferably, the dinitrogen-depleted anode stream **36** (recycling stream **36**) has a residual dinitrogen N₂ content greater than or equal to 10% by volume, more preferentially greater

than or equal to 20% by volume, typically between 20% and 40% by volume, for example between 20% and 30% by volume.

[0168] Finally, the dinitrogen-depleted anode stream **36** is injected into the ammonia gas stream **14**, where it is recycled to the fuel-cell unit.

[0169] According to a preferred embodiment, the method according to the invention does not comprise the step of using one or more hydrogen pumps. The method according to the invention is thus advantageous in that it eliminates the need for hydrogen pumps.

[0170] The plant **10** and the method according to the invention thus enable the production of an electric current **12** by means of a fuel-cell unit **30** supplied with ammonia. In addition, the plant **10** and the method described herein enable unreacted dihydrogen to be recycled back into the fuel cell.

[0171] The invention is based on the fact that the residual presence of dinitrogen N_{2} in the recycling stream **36** has little or no effect on the operation of the fuel-cell unit **30**. Dinitrogen N_{2} is a neutral gas for the fuel-cell unit **30**. At the operating temperatures of the fuel-cell unit **30**, the dinitrogen is inert and does not take part in any reaction that may occur inside the fuel-cell unit **30**. However, too much nitrogen in the fuel cell can reduce fuel cell performance by diluting the dihydrogen fuel too much when it comes into contact with the anode system **68**. The separation performance of the separation unit **102** enables a stream **36** to be generated whose stoichiometric proportions are as close as possible to the hydrogen/dinitrogen stoichiometry of the ammonia gas stream **14**, once its decomposition has taken place within anode system **68**. In this way, the feed stream **38** resulting from the mixture of ammonia gas stream **14** and dinitrogen-depleted recycled stream **36** is characterized by a hydrogen-nitrogen ratio very close to that of the ammonia gas stream **14**. In fact, the electrochemical performance of the fuel-cell system **30** is not affected by this recycling **36**.

[0172] In addition, the plant **10** and the method according to the invention significantly reduce the energy requirement associated with recycling dihydrogen H_{2} by allowing a significant proportion of the dinitrogen N_{2} recovered at the outlet of the fuel-cell unit **30** to be injected into the feed stream **38**.

[0173] In methods of the prior art, the recycling stream is essentially dihydrogen H_{2} . In particular, all the dinitrogen present in the anode gas stream has been separated so that the recycling stream is virtually dinitrogen-free. Separation and purification steps requiring more specific separation equipment and/or a high energy requirement are then necessary to achieve the required purity.

[0174] In the context of the invention, the residual presence of dinitrogen in the recycling stream **36** means that less complex separation equipment can be used. The lower degree of purity allowed by the method of the invention also makes it possible to significantly reduce the energy requirements associated with the aftertreatment of the anode gas stream **40** prior to its recycling.

[0175] A further advantage of the method and system is that excess heat emitted by the fuel-cell unit can be recovered and recycled. In particular, the anode gas stream **40** recovered at the outlet **76** of the fuel-cell unit **30** is cooled in at least two stages, on the one hand by the first heat exchanger system **46**, and on the other by the supplementary cooling module **84**. The first heat exchanger system **46** cools the anode gas stream **40** by heat exchange with the ammonia-rich feed stream **38** to an intermediate temperature of 200° C. to 500° C., preferably 300° C. to 450° C. Then, the supplementary cooling module **84** cools the pre-cooled anode gas stream **86** leaving the first heat exchanger system **46**. The supplementary cooling module **84** typically consists of a heat exchanger for recovering, preferably recovering and reusing, the heat from the pre-cooled anode stream **86**. This heat is recovered, for example, by feeding a steam generation system, a Rankine-type electrical generation system, or a heat distribution network.

[0176] When the separation module **102** comprises a membrane separation unit (as described above), it is possible that the dinitrogen gas stream **18** recovered at the outlet of the module **102** is under pressure. Preferably, in this case, the plant **10** according to the invention comprises, between

the separation module **102** and the plant **26**, a complementary system of turbines designed to generate a complementary electric current by expansion of the pressurized dinitrogen gas stream **18**.

[0177] The method and plant according to the invention are also advantageous in that they enable at least **80%** of the dihydrogen formed by decomposition of ammonia to be consumed in the fuel-cell unit **30**, preferably at least 90%, more preferentially at least 95%, typically at least 99% and advantageously all of the dihydrogen formed by decomposition of ammonia.

[0178] Last but not least, the system and method described in this invention are also advantageous in that they enable fuel that has not been consumed in the fuel-cell unit to be efficiently recycled.

Claims

1. A method for implementing a fuel-cell system, comprising the following steps: a) operating a fuel-cell unit comprising at least one anode system and at least one cathode system, said fuel-cell unit being continuously supplied with an ammonia-rich gas stream injected at the anode system and with a dioxygen-rich gas stream injected at the cathode system; b) recovering a dinitrogen-and dihydrogen-rich anode gas stream and a cathode gas stream at the outlet of the fuel-cell unit; c) cooling the anode gas stream and condensing the water present in the anode gas stream to form a cooled anode gas stream; d) separating the cooled anode gas stream into a dinitrogen gas stream and a dinitrogen-depleted anode stream; e) injecting at least part of the dinitrogen-depleted anode stream into the fuel-cell unit, in particular at the anode system, so as to recycle said at least part of the dinitrogen-depleted anode stream-into the fuel-cell unit, wherein the dinitrogen-depleted anode stream has a residual dinitrogen content greater than or equal to 5% by volume.
2. The method according to claim 1, wherein the dinitrogen-depleted anode stream has a dinitrogen content greater than 10% by volume, preferably greater than 20% by volume, more preferentially from 10% to 50% by volume, even more preferentially from 15% to 40% by volume, advantageously from 20 to 30% by volume.
3. The method according to claim 1, wherein said at least part of the dinitrogen-depleted anode stream is injected into the ammonia-rich gas stream to form an ammonia-rich feed stream, said feed stream then being injected into the fuel-cell unit, in particular at the anode system
4. The method according to claim 3, wherein the feed stream has a dihydrogen content greater than or equal to 15% volume, preferably greater than or equal to 40% volume, more preferentially from 40% to 50% volume.
5. The method according to claim 1, wherein step d) of separating the cooled anode gas stream is carried out by membrane separation, pressure swing adsorption, temperature swing adsorption, or pressure-temperature swing adsorption.
6. The method according to claim 5 further comprising between the steps c) and d) an intermediate step of compressing the cooled anode gas stream, preferably to a pressure greater than or equal to 10 bara, more preferentially greater than or equal to 20 bara, typically between 20 bara and 40 bara.
7. The method according to claim 1, wherein the anode gas stream is at least partially cooled in step c) by heat exchange with the ammonia-rich gas stream or feed stream.
7. The method according to claim 1, wherein the dioxygen-rich gas stream, before being introduced into the fuel-cell unit, is heated at least partially by heat exchange with the cathode gas stream.
9. The method according to claim 1, wherein it does not comprise a step of burning the anode gas stream, not even one of partial burning.
10. A fuel-cell plant comprising: a fuel-cell unit comprising an inlet for introducing an ammonia-rich gas stream, an inlet for introducing a dioxygen-rich gas stream, an outlet for recovering an anode gas stream, and an outlet for recovering a cathode gas stream, a cooling and condensing unit for cooling and drying the anode gas stream to form a cooled anode gas stream, a separation unit for separating the cooled anode gas stream into a dinitrogen gas stream and a dinitrogen-depleted

anode stream, said separation unit comprising a recovery outlet, said dinitrogen-depleted anode stream recovery outlet for recovering the dinitrogen-depleted anode stream being connected to the inlet for introducing the ammonia-rich gas stream into the fuel-cell unit, wherein the separation unit comprises at least one separation module selected from the group consisting of: a membrane separation unit, a pressure swing adsorption unit, a temperature swing adsorption unit, a pressure-temperature swing adsorption unit and any combination thereof.

11. The plant according to claim 10, wherein the separation unit further comprises a compression module, located upstream of said separation module.
